

Final Project

Food Ordering Web Application

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Dr. Nabeel Al-Mushasha

Submitted by:

Wejdan Saad Alzahrani	2240003203
Hind Hussein Shayan	2240002977
Hams awad almalki	2240003976
Lama Fareed Almoaidy	2240007804

Group (10)

Imam Abdulrahman Bin Faisal University

College of Business Administration

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<https://github.com/HindHussain/Food-Ordering-Website.git>

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Executive Summary

This project proposes implementing a Food Ordering Website that provides an online food ordering solution that enhances and facilitates the food ordering experience for users than previous traditional food ordering solutions that have limited menus and long wait times before receiving food or services from the restaurant. The website will include an interactive menu which allows users to scroll through different categories of food items and build a shopping cart to keep track of food items added/removed/modified to the cart, and once a user has completed his/her order, there will be a secure checkout page with order confirmation upon submission. The website was built using HTML, CSS, and JavaScript, will work on all devices including tablets, and will deliver an overall user-friendly/consistent web experience and provide restaurants with an efficient way to operate. The implementation of this website is an example of successful web development implementation via the Project Management process.

1.Introduction

The online food service industry is booming and demand has increased for digital platforms to facilitate seamless ordering of food. As a result, food vendors and restaurants increasingly have applications that are easy for customers to use and can accommodate a lot of users concurrently. This web application will allow customers to browse restaurant menus, choose food they'd like to eat and place food orders through an Internet connection will reduce potential manual errors, while allowing restaurants to be more efficient in the way they operate their business. The target audience for this application encompasses a wide range of customer bases, including individuals, families, and small groups that desire a convenient, fast, and dependable way to order food over the Internet.

Objectives of this Project:

- Create a fully developed, mobile responsive web application.
- Enable users to navigate easily and receive access to different devices.
- Application should be built with the use of HTML, CSS, and JavaScript
- effectively plan, develop and implement a solution for an actual business need.

2. System Analysis and Planning

For this project, the System Analysis & Planning phase will perform a thorough analysis of both the functional and non-functional requirements to ensure a well-designed, user-friendly, and accessible food ordering website that meets your business goals as well technical requirements.

2.1 Functional Requirements:

The website has a full-fledged smooth online food ordering service; it could include a dynamic menu of first-class food items with respective categories and related descriptions and prices (Cheeseburger – 20 SAR, Pepperoni Pizza – 30 SAR). Customers will have the ability to add products, as well as the ability change product quantity, and have a display of total price while they shop. In order to accomplish that task, you would need provide information in written form on an easy to read informal form located on our website where you supply your name, street address, and telephone number; however, you must complete the order process to confirm your purchase.

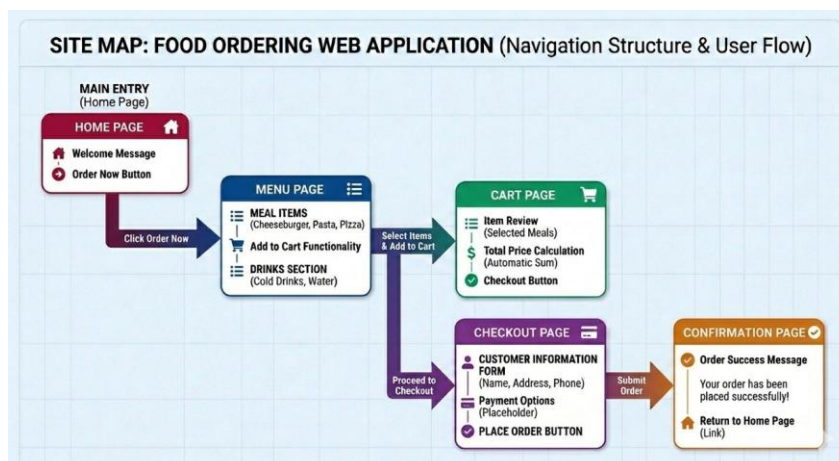
2.2 Non-Functional Requirements:

The whole site should work well on both mobile/desktop devices and offer a positive UX through consistent navigation, layout and professional Typography. The website's performance characteristics will support many concurrent users, will possess a high degree of scalability, and will have faster than average download speeds in order to maintain a high degree of user satisfaction.

User Roles:

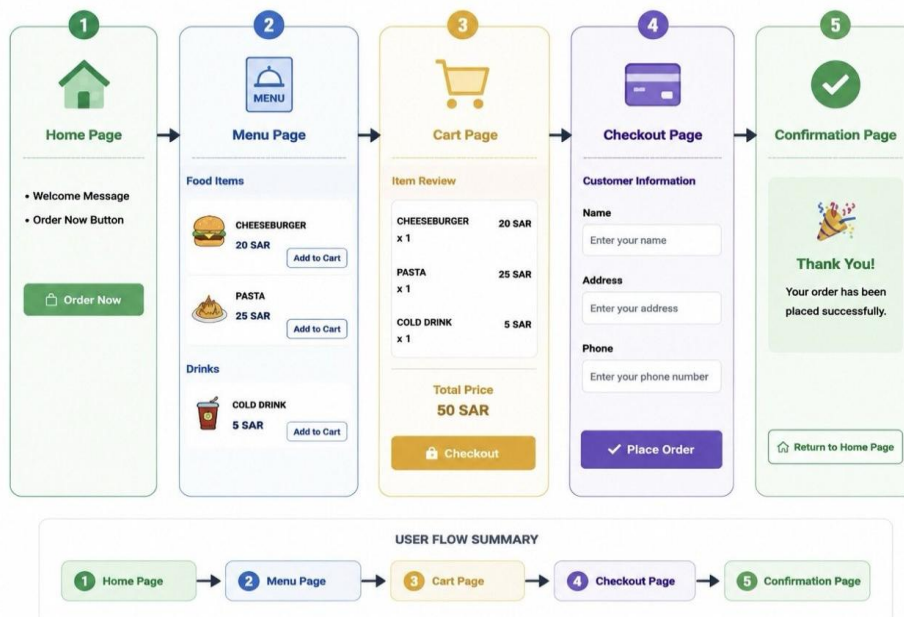
Customer: Will have access to the online menu, to manage their shopping cart, and to place their order for delivery.

Admin: Updates menu items, modifies prices, monitors order status, and ensures system operation with secure administrative access.



SITE MAP: FOOD ORDERING WEB APPLICATION

User Flow from Start to Order Completion



3. Core Business Functionality

The Food Ordering Site enables customers to order food online by browsing categories of food items using images, descriptions, and prices. Customers are able to add items to their dynamic shopping cart, to manage selected items, before they complete the validated checkout form and provide their customer details (i.e., name, address and phone number) for delivery. Efficient searching and filtering capabilities allow customers to quickly find food items they want. The user flow of the web site is defined as follows: Home >> Menu >> Cart >> Checkout >> Order Confirmation. JavaScript is used to manage dynamic cart updates, real-time calculations of item prices, and form validation to ensure that all essential checkout information has been entered correctly by customers before submitting their orders.

Three essential technical features of the web site:

- **Dynamic Lists of Products:** Food products can be grouped into categories, including burgers, pasta, and drinks, complete with corresponding pictures, descriptions, and prices.
- **Dynamic Shopping Cart:** Customers manage their selections and have their total order cost calculated automatically based on selected items.
- **Validating Checkout Forms:** Checkout forms capture the name, address, and telephone number of customers accurately before submitting the order form for processing.

- **Efficient Data Filtering and Searching:** Customers can narrow down the food items they wish to order using search and filter capabilities defined in the database.
- **JavaScript manages the interaction and updates of the shopping cart.** The shopping cart can include adding or removing items and updating the total cost in SAR.

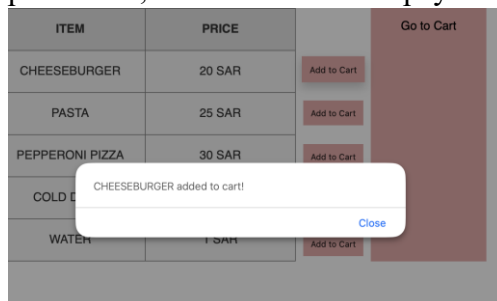
This website has met the project's objectives by delivering a working product that includes a responsive menu, interactive shopping cart, and secure checkout process. There was a lot of difficulty with the shopping cart's ability to be dynamic and have all the required validation associated with it. The main emphasis of the design portion of the challenge was to establish a sustainable level of congruity between the multiple platforms accessing the website. The key learnings from this implementation are that all projects must be tested and that each project requires efficient teamwork as well as appropriate documentation, among other things. The future improvements that could be added to projects are a secure payment option, improved search capability, and allowing users to create an account in order to monitor their orders.

4. System Testing and Validation

4.1 Testing and Validation

In this phase, rigorous testing was conducted to ensure the application's stability and performance. The following procedures were implemented:

- **Test Case Development:** Comprehensive test scenarios were designed to evaluate core application functions, including "Add to Cart" logic, order placement, and the end-to-end payment process.



- **Visual Documentation:** Each test result was documented with screenshots to verify that features operate correctly under different conditions and to provide evidence of successful execution.

- **Form Validation:** Systematic checks were performed on all user input forms (e.g., Shipping and Payment forms) to ensure data integrity and prevent errors during the data entry process.

The screenshot shows a checkout form with a dark red header. Below the header are three input fields labeled 'Name', 'Address', and 'Phone Number' (in Urdu). A white success message box in the center says 'Your order has been placed successfully!' with a 'Close' button. Below the form is a 'Place Order' button and a 'Terms and Conditions' link. At the bottom, it says '© 2020 All rights reserved'.

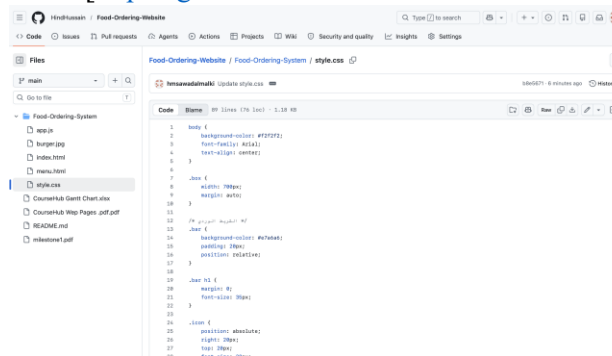
- **Issue Tracking and Resolution:** A dedicated log was maintained to document technical bugs or performance issues encountered during testing. Each issue was analyzed and resolved to enhance the overall user experience.



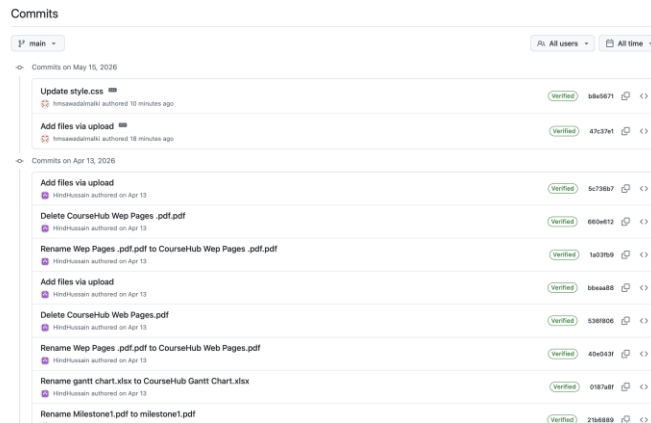
4.2 GitHub and Version Control

To manage the development process efficiently and maintain code integrity, GitHub was utilized as the primary version control system:

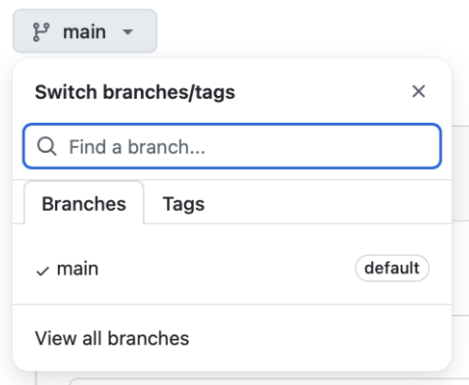
- **Project Repository:** The source code is hosted on a professional GitHub repository, ensuring a centralized and secure environment for the project files.
 - **Link:** [<https://github.com/HindHussain/Food-Ordering-Website>]



- **Commit History:** A consistent "Commit History" was maintained to track the evolution of the project. Screenshots of these logs are included to demonstrate the incremental updates and progress made throughout the development lifecycle.



- **Change Documentation:** Each update pushed to the repository was accompanied by a descriptive message, clarifying the specific modifications, bug fixes, or new features implemented in every version.
- **Branching Strategy:** An organized workflow was followed using a structured branching strategy:
 - **Main Branch:** Reserved for the final, stable version of the application.
 - **Feature Branches:** Used for developing and testing new components independently before merging them into the main codebase, ensuring that the primary project remains functional at all times.



4. System Design

4.1 Wireframes & Mockups

These mockups show the visual design of the Food Ordering Website. Each page was designed with a consistent pink header (#EFA5A6), a hamburger navigation menu, and Arial font to ensure a unified and cohesive user experience.

Figure 1: Home Page Mockup

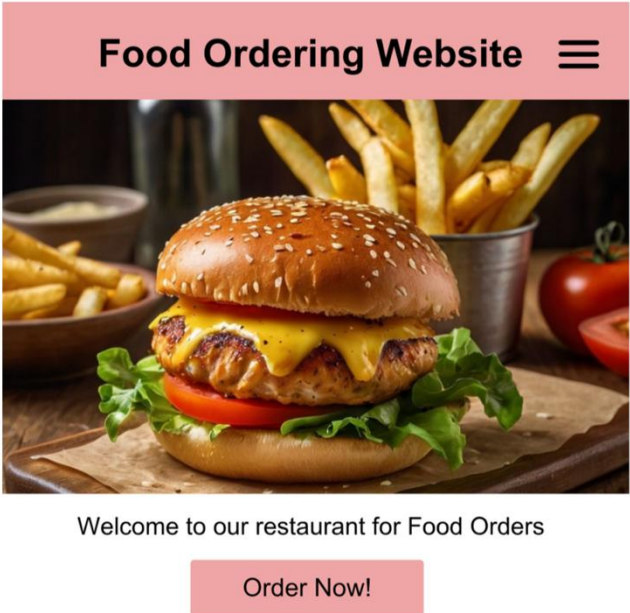


Figure 2: Menu Page Mockup

Menu 		
ITEM	PRICE	
CHEESEBURGER	20 SAR	Add to Cart
PASTA	25 SAR	Add to Cart
PEPPERONI PIZZA	30 SAR	Add to Cart
COLD DRINK	5 SAR	Add to Cart
WATER	1 SAR	Add to Cart

Figure 3: Cart Page Mockup

Your Cart

ITEM	PRICE
CHEESEBURGER * 1	20 SAR
COLD DRINK * 1	5 SAR
TOTAL	20 SAR

Checkout

Figure 4: Checkout Page Mockup

Checkout

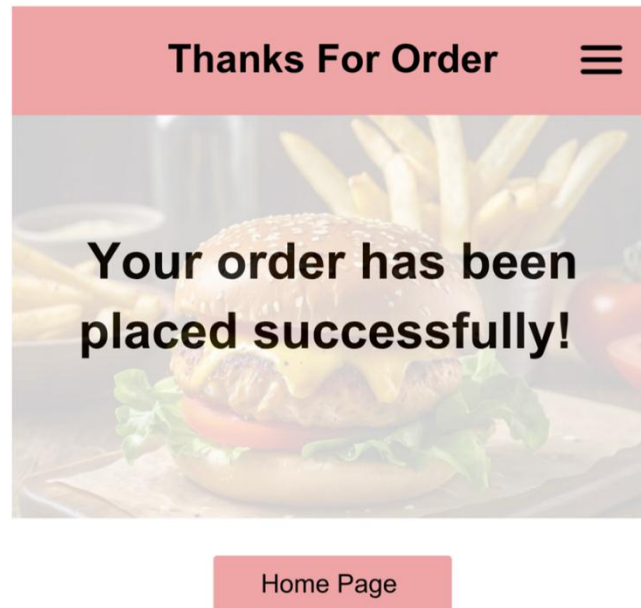
Name

Address

Phone

Place Order

Figure 5: Confirmation Page Mockup



4.2 UI/UX Design Decisions

With the Food Ordering Site, every effort has been made to keep the site looking very simple and/or easy to use. As such, there are certain design elements that were incorporated to enhance the overall user experience:

Color:

The primary colour used across the whole Website will be our pink colour #EFA5A6 for all Page Headers and Buttons to provide the user with a warm and inviting effect.

Font:

The font used throughout the entire Website will be Arial, but each Page will use a different size based on either the Heading (50pt bld.), Body (18pt) or Button (21pt to 32pt) sizes used throughout.

Navigation:

A Hamburger Menu (3 horizontal lines), located in the top right corner of each Page will allow the user to access all 5 Pages (Home, Menu, Cart, Checkout & Confirmation) by clicking on this button.

Layout:

All Pages will maintain the same overall layout, with the Pink Header located at the top of the Page, the Main Content is in the centre of the page and the Navigation Buttons at the bottom of the Page.

Color Scheme:

Element	Color Code
Header Background	#EFA5A6
Buttons	#EFA5A6
Table Header	#D3D3D3
Background	White
Text	Black

Typography:

Element	Font	Size	Style
Page Heading	Arial	50px	Bold
Body Text	Arial	18px	Regular
Menu Buttons	Arial	21px	Regular
Cart/Checkout/ Confirmation Buttons	Arial	32px	Regular

6. Implementation & Development

The Food Ordering Website was developed using three core technologies: **HTML, CSS, and JavaScript**. Each technology played a specific role in building the final product.

6.1 HTML Structure

To guarantee a well-organized, easy-to-read webpage, every page on this website followed semantic HTML element usage:

- The pink navigation bar at the top of every page is designated as the `<header>`
- The main page title is designated as the `<h1>`
- The hamburger navigation menu is designated as the `<nav>`
- The main content area is separated and organized by the use of the `<div>`

- Menu items and prices are displayed in table format on the Menu and Cart pages using the <table>
- The <button> element was used for every interactive button on all pages created.

6.2 CSS Styling

All style elements for the website have been created using CSS, and every HTML page has linked to the stylesheet, style.css to keep a consistent look and feel across the website's different pages. The following points demonstrate how CSS was applied:

- The pink background (header colour #EFA5A6) was applied for each of the headers with the class .header on all pages.
- Each hamburger menu was aligned along the top right corner of every page in the site using flexbox.
- The buttons (.btn-home, .btn-checkout, etc) incorporate a pink background, provide Arial typeface, and include rounded corners.
- The table header background colour of light grey (#D3D3D3) provides more visual separation from the table content.
- An animated (pulse) effect has been applied to the “Back to Home” button on the Confirmation page to grab user attention.

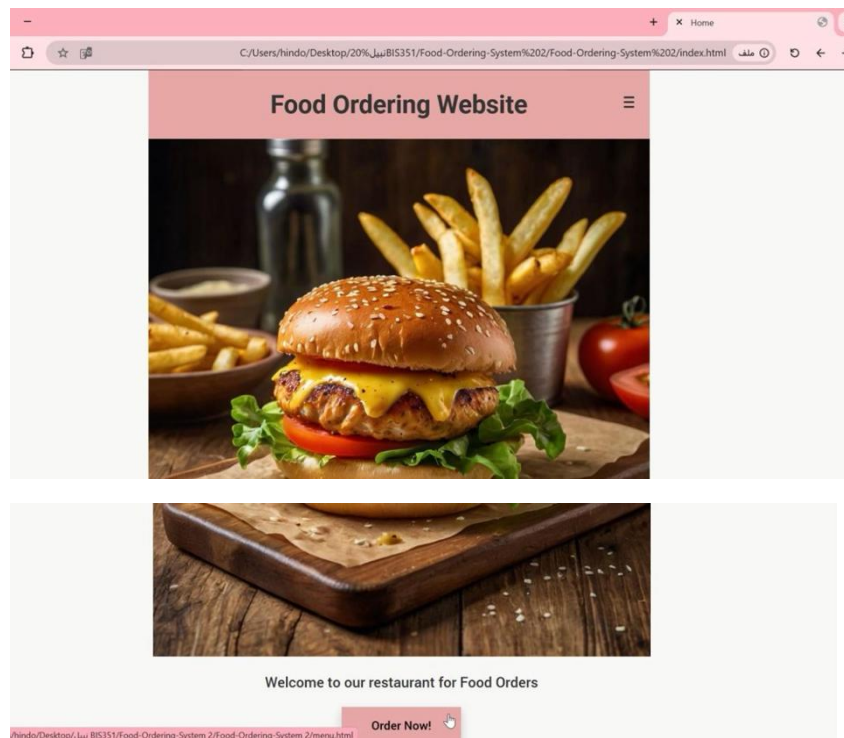
6.3 JavaScript

JavaScript was used to add interactive functionality across all pages of the website. The following features were implemented:

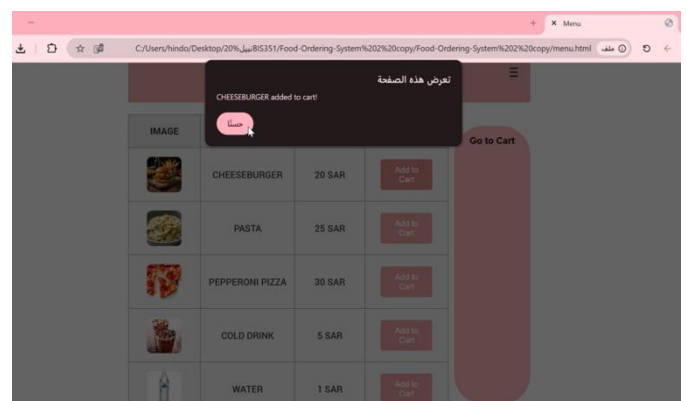
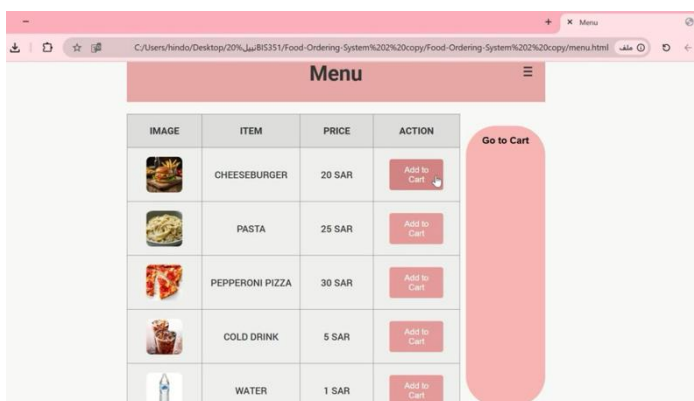
- All pages include a function that lets the user open/close hamburger menus via the three horizontal lines located thereon.
- Use of JavaScript on the Menu Page allows a user to click on a food item to gain access to an “Add To Cart” option.
- Use of JavaScript on the Cart Page allows a user to view items that have been added to the cart, calculate a total amount owed based on those selected items, and delete selected items.
- Use of JavaScript on the Checkout Page provides required Field Validation for use of the website prior to submitting for Order Placement. (Name, Address, Phone).

The following screenshots demonstrate the final implementation of the Food Ordering Website:

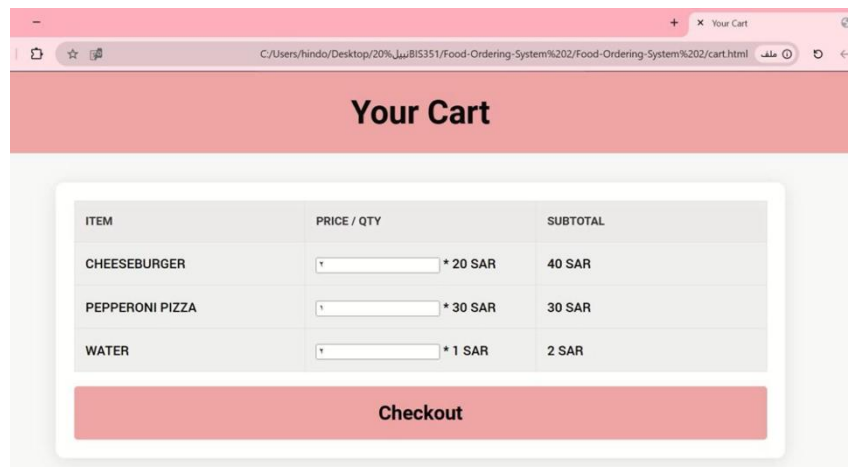
- **Figure 1: Home Page**



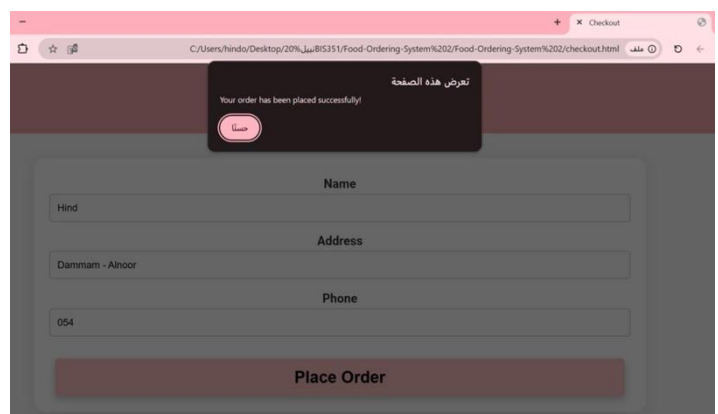
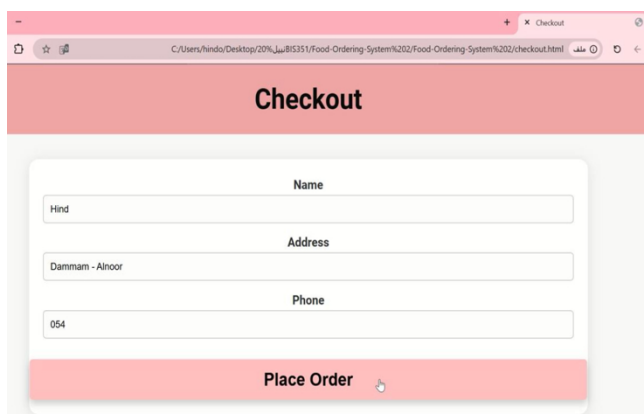
- **Figure 2: Menu Page**



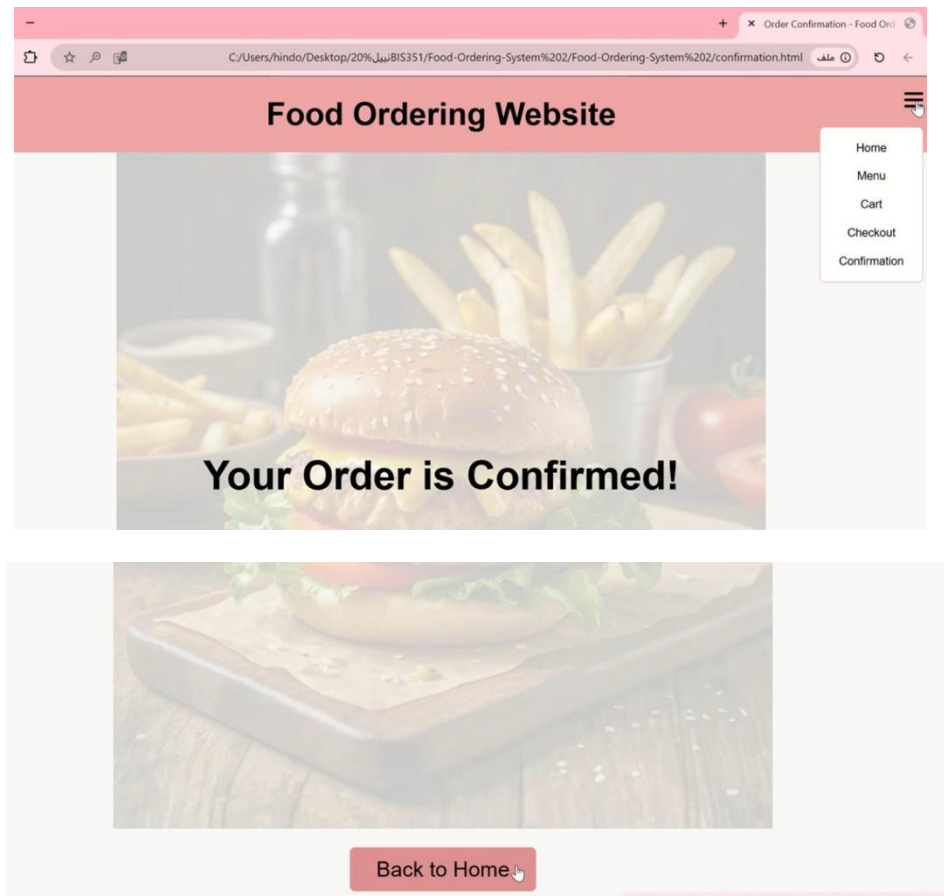
- **Figure 3: Cart Page**



- **Figure 4: Checkout Page**



- **Figure 5: Confirmation Page**



7. Challenges and Lessons Learned

- **Technical Issues:**

The majority of the complications arose from ensuring the synchronizing of my menu page selections sending to my cart view, since I was limited to using client side scripting alone without any persistent database.

- **Design Issues:**

Creating a responsive UI while maintaining the aesthetic looks of the food image in a checkout form that still works well on the smaller mobile screens.

- **Time Management:**

In order to stay within the timeline, I had to create a plan to code in a structured way with regards to which features should be developed first (e.g., cart and checkout validation) over secondary aesthetic features.

- **Future Enhancements:**

In future versions, we plan to implement a real time backend database (i.e., such as Firebase) so that user accounts have persistence, and also have the ability to track their order history.

8. Conclusion

The project produces a fully functional Food Ordering Site for customers to view menus, order product(s) and checkout. All of the requirements for the project have been met; menu is responsive, the shopping cart is interactive and the checkout form has been validated. This project clearly demonstrates the successful application of HTML, CSS, and JavaScript in creating a user friendly and efficient online ordering system. The development process also offered many lessons learned about creating dynamic elements to be integrated into a well-designed web site as well as making those sites usable from multiple devices. Key areas for future enhancements include adding a real-time search feature, developing an online payment gateway, and creating a user login system.

References

- ❖ W3Schools, 2026. *HTML Tutorial*. [online] Available at: <https://www.w3schools.com/html> (<https://www.w3schools.com/html/>)
- ❖ W3Schools, 2026. *CSS Tutorial*. [online] Available at: <https://www.w3schools.com/css> (<https://www.w3schools.com/css/>)
- ❖ W3Schools, 2026. *JavaScript Tutorial*. [online] Available at: <https://www.w3schools.com/js> (<https://www.w3schools.com/js/>)
- ❖ MDN Web Docs, 2026. Client-side Web Development. [online] Available at: https://developer.mozilla.org/en-US/docs/Learn/Front-end_web_developer
- ❖ Nielsen, J., 2012. *Mobile Usability*. Nielsen Norman Group.
- ❖ GitHub Docs, 2026. *About Git branches*. Available at: <https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/proposing-changes-to-your-work-with-pull-requests/about-branches>
- ❖ Chacon, S. and Straub, B., 2020. *Pro Git*. 2nd ed. Berkeley: Apress.
- ❖ W3Schools, 2024. *HTML, CSS and JavaScript Tutorials*. [online] Available at: <https://www.w3schools.com>
- ❖ Mozilla Developer Network (2024) *Web Development Documentation*. [online] Available at: <https://developer.mozilla.org>